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Research Title	Evaluating GLP-1 Receptor Agonists within Legalized Public Health Systems for Psychoactive Substance Dependence as Pharmacotherapy

Evaluating GLP-1 Receptor Agonists within Legalized Public Health Systems for Psychoactive Substance Dependence as Pharmacotherapy

Introduction: Psychoactive substance dependency remains a major global public health concern that severely affects individuals, families, communities, healthcare systems while presenting a critical challenge to society. Traditionally, the legislative and government bodies have responded to this crisis with punitive policies including custody and strict prohibition. Conversely, the public health community have started to comprehend substance dependence as a chronic medical condition and have encouraged for advocations toward pharmacotherapy and evidence-based medical treatment. Recently, Glucagon-like peptide-1 receptor agonists (GLP-1RA), with its primary role for treating diabetes and obesity, have attracted scientific interest due to their potential regulatory effects on appetite regulation, and addictive behaviours [1][2]. Emerging evidence suggests that these medications significantly influence drug and alcohol cravings. Furthermore, pharmaceutical advancements have pushed GLP-1RAs to develop its long-lasting effects and prolonging endocrine pancreatic sufficiency. With the existence of GLP-1RA drugs within legalized public health systems, it is possible to examine their clinical effectiveness, accessibility, and ethical implications, which ultimately contributes to the global transition from punitive enforcement to treatment-oriented public care.

Literature Review

GLP-1RA: Glucagon-like peptide-1 (GLP-1) is an incretin hormone that is secreted postprandially to stimulate insulin secretion from the pancreas, therefore results in monitoring of blood glucose homeostasis, significantly delaying digestion and suppressing the appetite via the metabolic centres by modulating releases of hormones and neurotransmitters [3]. Under the conditions of peripheral insulin resistance, the regulatory system may weaken and cause type 2 diabetes mellitus (T2DM). Glucagon-like peptide-1 receptor agonists (GLP-1RAs) such as exenatide, liraglutide, semaglutide and tirzepatide have developed to become highly effective treatments for T2DM.

The endogenous natural hormone GLP-1 exhibits metabolic clearance for approximately two minutes before fading, while the artificial drug resists degradation, extending the efficiency across days or weeks. This prolonged stability acts as a therapeutic breakthrough, with its seemingly safe pharmacodynamics that directly promotes weight loss and potentially reinforce healthier behavioural patterns. Specifically qualified and approved for obesity management, the World Health Organization included semaglutide (a prescription of GLP-1RA medication) to its Essential Medicines List for Diabetics in 2025 [4].

Dosage: The first synthetic GLP-1 (exenatide) that was approved in 2005 for T2DM still haven't demonstrated substantial issues and consistently maintained its glycaemic controls unless paired with other diabetic medications such as synthetic insulin. Initially introduced in as a twice-daily injection, have subsequently expanded its dosing options to include a once-

weekly formulations. Various risks and harms have been under careful investigation, but semaglutide has been notably correlated with reduced risks of myocardial infarction, Alzheimer's disease and cancer. Both formulations are still regularly prescribed today and haven't shown major issues [5]. GLP-1RA drugs have been approved for effective pharmaceutical therapies with findings that spoke up for slowing down the addicting effects of potential cravings. Multiple versions of the exenatide drug greatly reported the administration of subjects with nicotine-dependency or neurological disorders predisposed to risks of substance misuse, having been compared to contemporary pharmacotherapeutic drugs [2][6].

Preclinical Evidence & Current Gaps

Alcohol Dependence: Alcohol use disorder (AUD) is a medical condition in which controlled alcohol consumption and intense craving become unavoidable, and continued practice could lead to harmful consequences. Growing evidence indicates that GLP-1RAs may influence brain reward pathways and reduce intake of alcohol and craving. C.S. Hendershot et al. (2025) investigated the effects of once-weekly subcutaneous semaglutide on adults with AUD [7]. The randomized trial found no impacts on alcohol consumed per calendar year, but a significant reduction on number of drinks taken per day with greater reductions in heavy drinking over time [7]. Despite certainty and wide acceptations, there is limited large-scale clinical trials, and the evidence are far too ascertain when regarding long-term outcomes.

Nicotine Dependence: Tobacco use disorder or nicotine dependence is a chronic disease that is characterized by a compulsive craving for nicotine. Continued using of tobacco products lead to loss of control over intaking and leads to negative health effects and shortening of lifespan. S. Lee et al. (2024) assessed questionnaire of smoking urges (QSU) and found that after six weeks of intervening post-nicotine cessation with a combination of GLP-1RAs, the results favoured lower urges to crave for nicotine and to smoke [8]. C. S. Hendershot et al. (2025) also found that the semaglutide treatment caused the reduction of cigarettes taken per day by a few samples [7]. Present studies and clinical data justify that GLP-1RAs may be potential treatments for people with NUD, and it is evident that GLP-1RA has a promising relationship with behavioural patterns and cravings for nicotine or tobacco. However, many trials still encourage for future research on the effects of GLP-1RAs on smoking behaviours and other drug abuses.

Opioid Dependence: Opioid use disorder (OUD) is characterized by intense cravings and the compulsive using of opioids. Examples of opioids include substances such as fentanyl, methadone, heroin and other morphine-like agents that can cause common toxicities such as overdosing and psychological deterioration. Human and animal studies of GLP-1RAs (exenatide and liraglutide) being tested to examine the impact on opioid craving behaviours resulted in the conclusion of GLP-1RA drugs being linked to the reduction of seeking to use and take heroin and other opioids [9]. Previous studies have also highlighted that GLP-1RAs may impact the spinal pathway and intervene the mesolimbic pathway, which may interact with more parts of the brain and neuroinflammatory parts of the body. Much of the clinical

trials are still limited and small-scale, which request for a much larger and iterative approach for the future of GLP-1RAs for pharmacotherapy.

Research Questions:

This study aims to evaluate clinical efficiency and ethical viability, as well as reliability of integrating GLP-1 receptor agonist drugs as an alternative to concurrent pharmacotherapeutic drugs and punitive policies.

Specifically, the study focuses on answering these topic-related questions:

Socio-Economy

1. How does transitioning from punitive policies against psychoactive dependence to GLP-1 assisted public health treatment impact overall retention and relapse rates for individuals with substance use disorders (SUD)?
2. What kind of systemic or socioeconomic barriers exist when scaling up the distribution of GLP-1RA drugs to health clinics for substance dependence?

Clinical Effect

3. To what extent do GLP-1RA drugs reduce neural cravings and consumption patterns in individuals transitioning away from psychoactive substances?
4. How do GLP-1RA therapies compare to concurrent pharmacotherapies such as buprenorphine and naltrexone within a public health setting?

Ethical & Legal Impact

5. What is the long-term benefit analysis of deploying GLP-1 receptor agonists as a frontline treatment for substance dependence regarding both public health and legal criminal justice enforcement?

These questions are clear, contemporary and focused, as it identifies the causes and uses measurable effects whilst also considering a clinical population. Additionally, the questions are highly effective and timely, as it compares clinical medicine with public health policy, highlighting the importance of GLP-1RA that aligns with modern scientific interests.

Proposed Methodology: A 6-month long project will be carried out to use mixed-methods where anonymity is maintained for all participants. The study is based on the existing GLP-1RAs semaglutide, liraglutide and related agonists. However, the methodology will not assume that GLP-1RAs have already established replacement for concurrent medications for psychoactive dependency. Instead, the existing pharmacotherapies such as buprenorphine and naltrexone will be examined whether considering the potential role of GLP-1RA coming to effect in the potential future by carefully assessing the sufficient evidence.

The detailed methodology goes as followed:

1. **Review & Data Collecting:** A meticulous review with detailed data analysis of ongoing clinical trials evaluating GLP-1 receptor agonists for SUD, specifically

opioid, nicotine and alcohol related disorders will be implemented. Comparing available evidence concerning GLP-1 pharmacotherapies including semaglutide or tirzepatide can be collected through multiple biomedical databases such as Clinicaltrials, PubMed and Cochrane Library. Sample sizes, participants' characteristics, GLP-1RA type and their given dosage and comparator treatment will also be assessed. Sufficiently comparable studies and findings will be used to design a quality synthesis for a complete literature review.

2. **Data Analysis:** Following the data collection, a careful analysis will be conducted to assess the mean differences, demographics and variables. The data, particularly reduction of substance craving and consumption will be carefully observed. The data analysis will consider the key differences in effectivity of treatments, safety of participants, pharmacodynamics and pharmacokinetics of GLP-1RAs, and accessibility of patients. Studies failing to meet the inclusive criteria or irrelevant findings will be excluded from the final academic synthesis.

3. **Qualitative Approach:** Small scale interviews with healthcare experts, addiction treatment clinicians and pharmacists aiming to assess twenty participants *(n = 20)*, where five participants *(n = 5)* will be selected from each of the four groups where subjects possess lived experience of substance dependency will be carried out. The interviews will explore potential benefits and clinical behaviours, risk patterns and the examination of buprenorphine or naltrexone with regulation and ethical considerations. The results of the interviews will be carefully analysed thematically to identify potential patterns and significant differences among the participants. The assessment is to acknowledge the existing evidence and emerging development before sufficient evidence is available for use.

4. **Quantitative Approach:** A structured questionnaire will be used to collect quantitative data on patients' perceptions of GLP-1RAs for treatment of SUD. A sample of respondents *(n = 100)* will be assessed, preferably people with relevant treatment experience of substance-dependency related disorders. The questionnaire will use a five-point Likert scale ranging from 1 = *'Strongly Disagree'* and 5 = *'Strongly Agree'* to measure the expectation of effectiveness in reducing substance cravings, willingness to support public-health clinics, and acceptance of replacing concurrent pharmacotherapies with GLP-1RA treatment. This approach will measure the ethical and socio-economic barriers among a common related participant category, and the results will be used alongside literature review and qualitative interviews to find the potential role of GLP-1RAs in treatment for SUD.

5. **Finalizing:** The findings from the literature review, and the results of the qualitative interviews and quantitative questionnaire will be compared to identify shared understandings, incompatible disagreements and uncertainty between them. To consider whether the current evidence supports GLP-1RA drugs as a therapeutic solution, the study will conclude its findings for effectivity in treatment, accessibility for patients and ethical liability.

Timeline

- **Month 1:** Setting clear objectives, ethical planning, preparing instruments
- **Month 2:** Database review, data collection
- **Month 3:** Conducting synthesis
- **Month 4:** Begin interviews, questionnaire
- **Month 5:** Mixed-methods data analysis
- **Month 6:** Check references, revise findings, create final report and manuscript

15 **Conclusion:** To conclude, this study will evaluate the emerging potential of GLP-1 receptor agonists for psychoactive dependence by using clinical evidence and public health perspectives. Ultimately, it will assess the effectiveness, safety, accessibility and ethical 24 measures of GLP-1 receptor agonists by identifying whether current evidence supports the future for public-health treatments with GLP-1RAs as pharmacotherapeutics.

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